PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Technology-Assisted Depression Screening Tools for Cancer Patients: A Systematic Review Protocol
AUTHORS	Lazaro-Escudero, Maria; Burgos-Cardona, Camila; Acevedo-Fernández, Karina; Castro, EM

VERSION 1 – REVIEW

REVIEWER	Ahmad S. Alzahrani a) Neuroscience Department, King Faisal Specialist Hospital and Research Center, Jeddah, Saudi Arabia b) College of Medicine, Alfaisal University, Riyadh, Kingdom of Saudi Arabia
REVIEW RETURNED	06-Sep-2020

GENERAL COMMENTS	Dear authors, Thank you for your well-written manuscript. The aim of the systematic review is important given the new era of technology-assisted tools in aiding medical assessment and intervention. Here are my comments and suggestions for your study: • Page 9, line 131: Inclusion criteria included “Studies evaluating the effectiveness, efficacy, feasibility and/or acceptance of the technology-integrated depression screening tool” would be better to define the measures used for evaluating the effectiveness, efficacy, feasibility. i.e., will you include both quantitative and qualitative methods of these measures? Will you exclude articles that did not define the criteria for effectiveness, efficacy, and feasibility (for example, studies that depend on only qualitative assessment). • Given the measures of efficacy and effectiveness, I would suggest to include only studies that confirmed the diagnosis of depression in the screening process (e.g., by interview or diagnostic instrument). Thank you.

REVIEWER	Doron Gothelf Tel Aviv University
REVIEW RETURNED	11-Sep-2020

GENERAL COMMENTS	This is timely and needed meta-analysis of an important topic. I suggest to add to the parameters pooled from each study the following parameters: percent of 'relapse' patients, if the study is cross-sectional or longitudinal (if longitudinal record the exact time points), if the scale yield one score or several factors and if cutoff values calculated or provided. The last point is especially important. In my view, there are many scales used for measuring

	depression in individuals with cancer but guidelines of implementation of systematic clinical screening (e.g,, valid cutoff values) are lacking.

REVIEWER	Erin K. Ferenchick Columbia University Irving Medical Center Center Family and Community Medicine New York, New York USA
REVIEW RETURNED	13-Nov-2020

GENERAL COMMENTS	Dear Authors, Thank you for this very thoughtful manuscript. You clearly highlight the under-recognition of depression in this specific patient population and your proposed scope of research will make an important contribution to identifying new approaches to depression screening. Below please find a few comments for your consideration: 1) Please indicate the time period (Month, Year - Month, Year) which will be covered in your review. 2) Clarify the model(s) of service delivery that will be explored. Have parameters been set for the review? For example, does it only include screening in oncology practices, or does it include primary care settings for cancer patients too? Does it include screening in the inpatient and/or outpatient setting? I see some variation in the language throughout the document with reference to oncology practices, government hospitals and cancer care systems. 3) What is actually meant when you talk about "IT devices" as well as "technology delivery model?" Does this reference the actual hardware (e.g., tablets, computers, mobile phones) or do you mean the IT platform (i.e,, the specific e-screening model)? It may be helpful to orient the reader and clearly define some of these terms and provide examples. 4) Consider clarifying the language used in your objectives. As your goal, do you want to identify the "most common" as you state or rather the most effective/efficient IT solution? 5) In the exclusion criteria, you note that you will not include studies incorporating the use of IT devices to facilitate baseline and follow-up assessment. Screening tools are often used as measurement tools so it would be good to better understand the rationale for this decision. Best of luck with your submission.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1

Thank you for your well-written manuscript. The aim of the systematic review is important given the

new era of technology-assisted tools in aiding medical assessment and intervention. Here are my comments and suggestions for your study:

1) Page 9, line 131: Inclusion criteria included “Studies evaluating the effectiveness, efficacy, feasibility and/or acceptance of the technology-integrated depression screening tool” would be better to define the measures used for evaluating the effectiveness, efficacy, feasibility. i.e., will you include both quantitative and qualitative methods of these measures? Will you exclude articles that did not define the criteria for effectiveness, efficacy, and feasibility (for example, studies that depend on only qualitative assessment).

2) Given the measures of efficacy and effectiveness, I would suggest to include only studies that confirmed the diagnosis of depression in the screening process (e.g., by interview or diagnostic instrument).

Response to reviewer #1: Thank you for your suggestions and for highlighting the importance of this study. Yes, we will exclude articles that do not define the criteria for effectiveness, efficacy and/or feasibility. We will not include articles using only qualitative assessments. We clarified this in the manuscript (Exclusion criteria section, page 6). We are not limiting the literature search to only studies that confirmed the diagnosis of depression because our main goal is identifying and describing the technology used for depression screening. However, based on your comment, we will document those studies that include the confirmation of the depression diagnosis. We have added this description to our Data items section of extracted data items (page 8).

Reviewer 2

This is timely and needed meta-analysis of an important topic.

1) I suggest to add to the parameters pooled from each study the following parameters: percent of 'relapse' patients, if the study is cross-sectional or longitudinal (if longitudinal record the exact time points), if the scale yield one score or several factors and if cutoff values calculated or provided. The last point is especially important. In my view, there are many scales used for measuring depression in individuals with cancer but guidelines of implementation of systematic clinical screening (e.g,, valid cutoff values) are lacking.

Response to reviewer #2: Thank you for your recommendation. We have included your suggestions on the Data Items section as follows: The following parameters will be pooled from each study selected: percent of ‘relapse patients’, study design (including exact time points when longitudinal), scale scoring criteria (including cut-off values) (page 9).

Reviewer 3

Thank you for this very thoughtful manuscript. You clearly highlight the under-recognition of depression in this specific patient population and your proposed scope of research will make an important contribution to identifying new approaches to depression screening.

Below please find a few comments for your consideration:

1) Please indicate the time period (Month, Year - Month, Year) which will be covered in your review.

2) Clarify the model(s) of service delivery that will be explored. Have parameters been set for the review? For example, does it only include screening in oncology practices, or does it include primary care settings for cancer patients too? Does it include screening in the inpatient and/or outpatient

setting? I see some variation in the language throughout the document with reference to oncology practices, government hospitals and cancer care systems.

3) What is actually meant when you talk about "IT devices" as well as "technology delivery model?" Does this reference the actual hardware (e.g., tablets, computers, mobile phones) or do you mean the IT platform (i.e,, the specific e-screening model)? It may be helpful to orient the reader and clearly define some of these terms and provide examples.

4) Consider clarifying the language used in your objectives. As your goal, do you want to identify the "most common" as you state or rather the most effective/efficient IT solution?

5) In the exclusion criteria, you note that you will not include studies incorporating the use of IT devices to facilitate baseline and follow-up assessment. Screening tools are often used as measurement tools so it would be good to better understand the rationale for this decision.

Response to reviewer #3: Thank you for acknowledging the value of this study. We have included the time period covered in our review on the manuscript (Search strategy section; page 7). We will be including all published articles up to December 2020. We will include all types of service delivery models in our review, including inpatient/outpatient settings, as well as oncology practices and primary care settings, thus, the reason why we did not include this parameter as criteria for inclusion/exclusion. The language variation observed is due to the terminology used in the reviewed literature (cited in the Rationale). In the Data Items section (second bullet; page 8) and Sample Table 2 (Second column; page 10) we had stated that we would extract study site information. We have now changed the term and used: “Model of service delivery type (oncology practice, primary care setting, inpatient setting, etc.)” to address the types of settings that will be explored and also maintain the same language regarding type of service delivery model. We also provided examples of types of IT devices (computers, tablets, mobile phones, etc.) and delivery models of IT (internet browser, type of platform, application software etc.) to better orient the reader on these terms (Objectives section; page 5). With this review our intention is to identify and describe the technology used for depression screening. We acknowledge that this is an emerging field, and thus, probably not ready for conducting a metanalysis to determine efficacy or effectiveness of technology-delivered depression screening tools. We will include studies that carried out an assessment of the IT device itself. We don’t want to include articles that use IT devices to measure intervention effects since the efficacy, effectiveness or feasibility of the device is not assessed. We decided to remove this exclusion criteria because we recognize it can be confusing and it is actually addressed in the followed exclusion criteria (“studies using technology devices to deliver behavioral interventions to treat depression”; second bullet; page 6).

VERSION 2 – REVIEW

REVIEWER	Ahmad Saad M Alzahrani * King Faisal Specialist Hospital and Research Center, Jeddah, Saudi Arabia. * College of Medicine, Alfaisal University, Riyadh, Saudi Arabia
REVIEW RETURNED	27-Dec-2020

GENERAL COMMENTS	Previous comments were covered in this revised version of the manuscript. Thank you.

REVIEWER	Doron Gothelf

	Sheba Medical Center
REVIEW RETURNED	26-Dec-2020

GENERAL COMMENTS	The authors have revised the manuscript and addressed well the points I raised.

REVIEWER	Erin K. Ferenchick Columbia University, Center for Family and Community Medicine (New York, USA)
REVIEW RETURNED	04-Jan-2021

GENERAL COMMENTS	Dear Authors, Thank you again for your important contribution to this scope of work. The recent edits are well noted, particularly the clarification of how the term "IT" is being used in the context of the review. I also take note of the clarifications regarding the "model of service delivery type." The manuscript references "oncological practice" as the point of care of interest but it appears that your inclusion criteria includes different sites where cancer-patients receive health services (i.e., primary care and in-patient). The effectiveness, feasibility, efficacy and acceptance of implementing these IT solutions may be quite different across these service delivery models given different levels of resources and specific enabling/hindering factors so this must be unpacked further in your analysis and discussion of the review. All the best in your next steps. Regards, Erin Ferenchick